Matthew Zeng
mz22353
March 31, 2026

COE 311K Midterm Project

INTRODUCTION

By fitting and interpolating the data with three curves, each coded from scratch, this report models and analyzes the US GDP growth rates from Q1 of 2010 to Q4 of 2023. The data points are sourced from US Bureau of Economic Analysis (BEA), which represent the seasonally adjusted annual rate (SAAR) of real growth of GDP in percent. The data covers 20 quarters over the 14 years, along with an additional analysis of the COVID-19 impact of 2020. The three methods for curve fitting and interpolation are: (**1) natural cubic spline interpolation**, which passes through each point while maintaining C^2 continuity; **(2) degree-4 polynomial fitting**, which estimates the global trend with least squares normal equations; and **(3) linear least squares regression**, used for the subset that excludes the COVID-19 quarters. Combined, these methods compare the differences between interpolation and smoothing when real considerations.

Matthew Zeng
mz22353
March 31, 2026

PART A — CUBIC SPLINE INTERPOLATION

A1. SYSTEM OF EQUATIONS

Given that we have $n = 20$ data points $(x_0, y_0), (x_1, y_1), \ldots, (x_i, y_i)$, there are $n - 1 = 19$ segmented cubic polynomials for each interval $[x_i, x_{i+1}]$:

$$S_i(t) = a_i + b_i(t - x_i) + c_i(t - x_i)^2 + d_i(t - x_i)^3.$$

There are 4 coefficients, meaning we have **4 × 19 = 76** unknowns:

$$S_i(x_i) = y_i, S_i(x_{i+1}) = y_{i+1} \rightarrow 2 \times 19 = 38\, unknowns - (C^0\ continuity\ at\ interior\ knots)$$

$$S_i'(x_{i+1}) = S_{i+1}'(x_{i+1}) \rightarrow 18\, unknowns - (C^1\ continuity\ at\ interior\ knots)$$

$$S_i''(x_{i+1}) = S_{i+1}''(x_{i+1}) \rightarrow 18\, unknowns - (C^2\ continuity\ at\ interior\ knots)$$

$$S_0''(x_0) = 0,\ S_{18}''(x_{19}) = 0 \rightarrow 2\, unknowns - (Natural\ boundary\ conditions)$$

A2. EVALUATION OVER ALL 55 QUARTERS

Using the data from 20 quarters, the spline can be evaluated at each quarter from index 1 (2010 Q1) to index 55 (2023 Q4), generating the plot below.

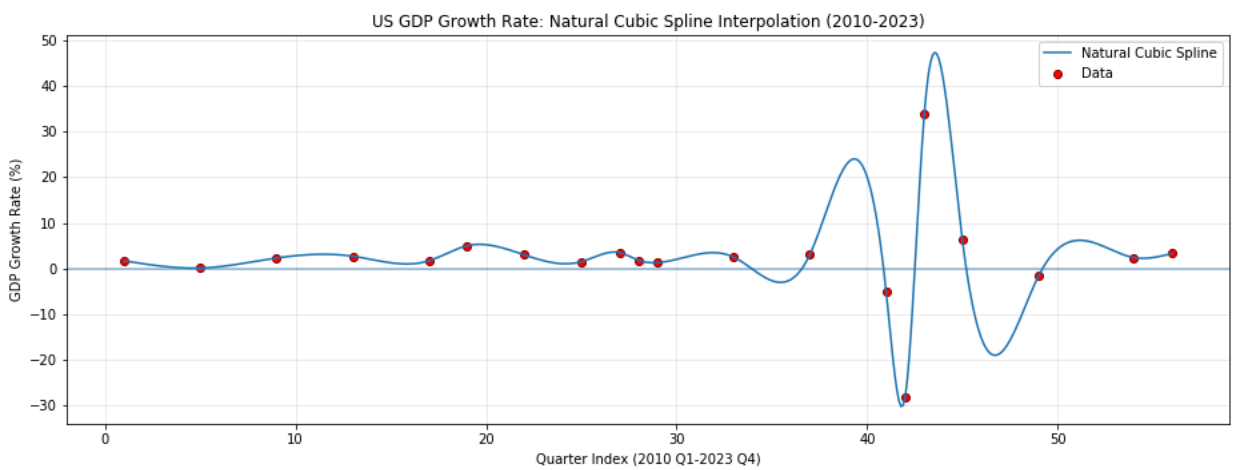

Figure 1. Natural cubic spline evaluated over all 55 quarters (2010 Q1 - 2023 Q4). Red dots are provided data points.

The curve is mostly smooth throughout the entire 14-year range. Outside of the COVID-19 years (2020-2021), the curve is quite even among the data, suggesting slight (1-3%) growth. Inside of the COVID-19 region (indices 41-49), there are very stark changes that cause overshots, even in near the outliers. These drastic oscillations are due to the large second derivative moments near the -28.1% and 33.8% knots.

A3. RUNGE PHENOMENON AND ALTERNATIVES

Matthew Zeng
mz22353
March 31, 2026

Runge's phenomenon is when the interpolating polynomial function results in high offsets or oscillations after being forced to pass exactly through extreme values. For this dataset, forcing the cubic spline through -28.1% and 33.8% causes the system to create large moments at those points. Due to C^2 continuity coupling between each cubic interval, those moments propagate through the spline, resulting in overestimating and underestimating the data.

We desire exact interpolation when the data consistently models the analyzed process. In our analysis, the COVID-19 quarters represent an inconsistency due to a special, unique circumstance that doesn't accurately represent the natural, long-term GDP growth. Forcing the interpolation through those extreme points inserts random oscillations without any economic significance. These uncertainties cause the spline to inaccurately interpolate surrounding quarters.

For this dataset, we have two alternatives that may better represent the model: smoothing splines and weighted least squares. **Smoothing splines** can minimize overfitting with

$$\Sigma_i (y_i - S(x_i))^2 + \lambda \int (S''(t))^2 dt$$

where λ regulates the fitting by allowing the spline to not exactly pass through certain extreme points. By trading off extreme interpolation accuracy, the spline becomes less sensitive to outliers and smoothens out. **Weighted least squares** method assigns a weight w_i to each point, and setting $w_i = 0$ for the COVID-19 quarters minimizes their impact while keeping them in the data. Both alternatives would generate a smoother interpolation spline with more accurate interpretations, making the COVID-19 quarters a more transient and temporary event.

Matthew Zeng
mz22353
March 31, 2026

PART B — POLYNOMIAL & LEAST SQUARES COMPARISON

B1. DEGREE-4 POLYNOMIAL FIT

Through normal equations

$$(V^T V)c = V^T y$$

where V is the Vandermonde matrix with

$$V[i][j] = t_i^j,$$

we fit the degree-4 polynomial

$$p(t) = c_0 + c_1 t + c_2 t^2 + c_3 t^3 + c_4 t^4.$$

We first centered the quarter indices to decrease the condition number of the normal equations matrix with

$$\hat{t} = (t - \bar{t})/\sigma.$$

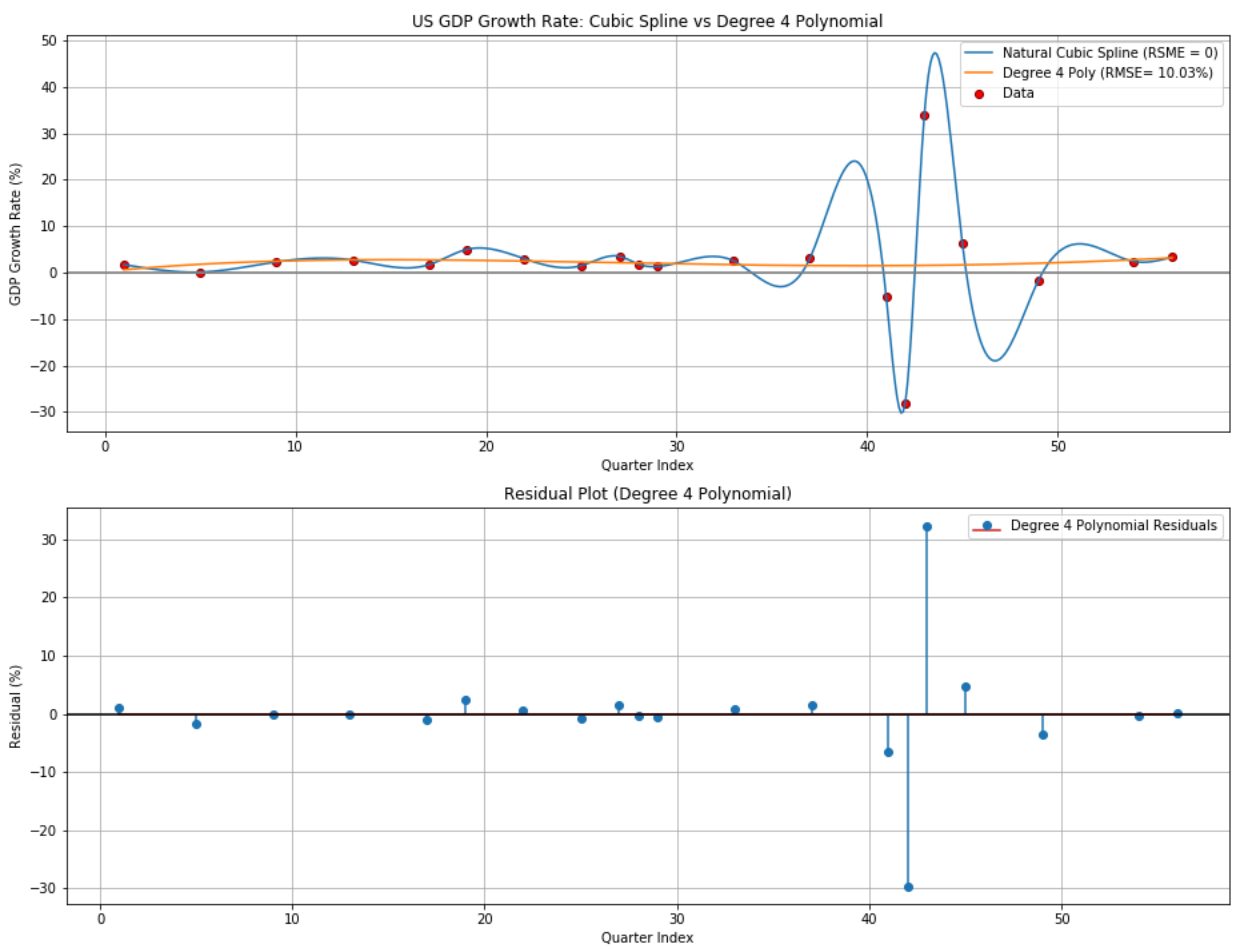

Figure 2. Top: Cubic spline vs degree-4 polynomial fit; Bottom: Degree-4 polynomial fit residual plot; RSME = 10.03%

Matthew Zeng
mz22353
March 31, 2026

The degree-4 polynomial has an RMSE of 10.03% while the cubic spline has an RMSE of 0% across the 20 data points. In the residual plot, the large errors are mostly gathered during the COVID-19 quarters, showing the overall interpolation from the polynomial due to the curve being affected by each value and the local interpolation from the cubic spline due to the curve being affected by surrounding values. The spline replicates the exact values of the data points while dramatically varying between the COVID-19 values, making it optimal for analyzing brief, specific time frames. The polynomial, however, smooths out around outliers while inaccurately reporting the exact values, making it optimal for long-term analysis across a broader time frame.

B2. LEAST SQUARES FIT (NO COVID-19)

We can exclude the COVID-19 quarters (2020 Q1 - 2021 Q1) and analyze the rest of the data with the linear model

$$y(t) = b_0 + b_1 t$$

with the normal equations

$$A^T A b = A^T y$$
$$A = [1 \mid t].$$

Since the COVID-19 quarters contribute sudden, noneconomic data, we can analyze the data without the corresponding values for a more accurate and representative interpretation.

Matthew Zeng
mz22353
March 31, 2026

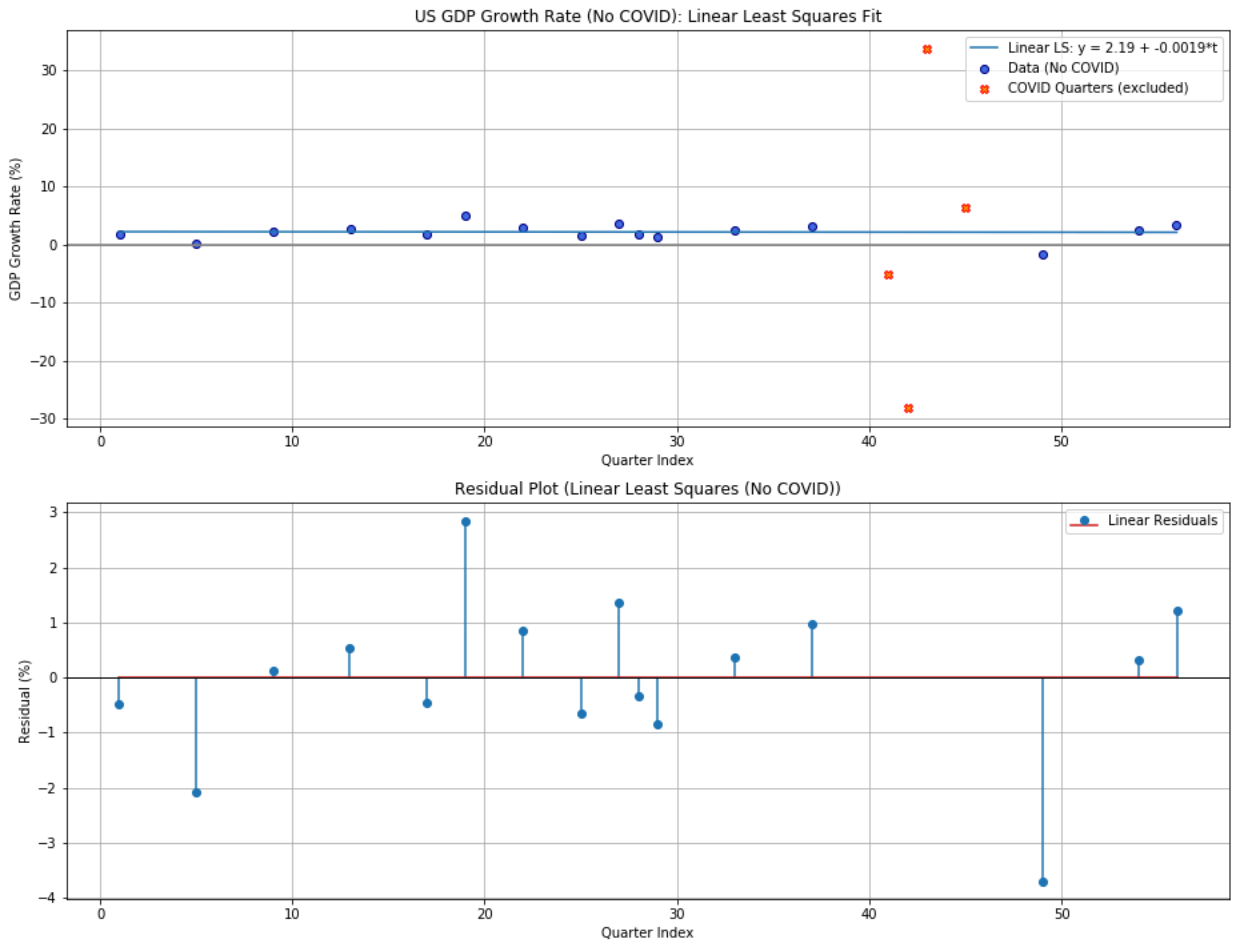

Figure 3. Top: Linear least squares fit without COVID-19 quarters (orange X points); Bottom: Linear fit residual plot; RMSE = 1.44%

The linear model is represented by

$$y(t) = 2.1943 - 0.0019t$$

The slope is -0.0019% per quarter, which translates to -0.0076% per year, but these are all essentially zero, meaning that there are no significant economic trends outside of the pandemic regarding US GDP growth. By maintaining around a 2% GDP growth rate, we see that the US has quite a stable economy, ignoring COVID-19. For a linear fit to make sense, it requires two assumptions: **(1) growth maintains the same rate over time**, meaning the trend follows a fixed constant, and **(2) each quarter is independent from each other**, as one quarter should not affect the growth of the next in a simple linear fit. From the 2014 Q3 and 2011 Q1 in the residual plot, we see outlying residuals that possibly challenge assumption (2). However, a linear model still makes reasonable sense and can help with initial analysis despite requiring more complicated models for a full, in-depth analysis.

Matthew Zeng
mz22353
March 31, 2026

PART C — METHOD JUSTIFICATION

Despite the COVID-19 impact, the **natural cubic spline** provides the best interpolation method for a policymaker who needs to interpolate GDP growth for a quarter due to three reasons:

(1) Smoothness: The spline forces a C^2 continuity condition everywhere. All intervals are only affected by their surrounding knots, so the curve is not as affected by values far from the pandemic. Although the degree-4 polynomial has C^∞ continuity due to it being one entire equation, it's affected by all data points at the same time, making it difficult to accurately interpolate at extreme intervals.

(2) Accuracy: The spline has RMSE = 0% while the degree-4 polynomial has RMSE = 10.03%. Because the spline exactly passes through each data point, its interpolation is technically better based on verified data points. Although the polynomial is less affected by extreme values, we don't have enough information to make the conjecture that it's more accurate.

(3) Sensitivity to Outliers: Due to Runge's phenomenon, the spline heavily overshoots the COVID-19 quarters. During those intervals, the natural cubic spline is inaccurate, and this is the main caveat to the cubic spline for this dataset

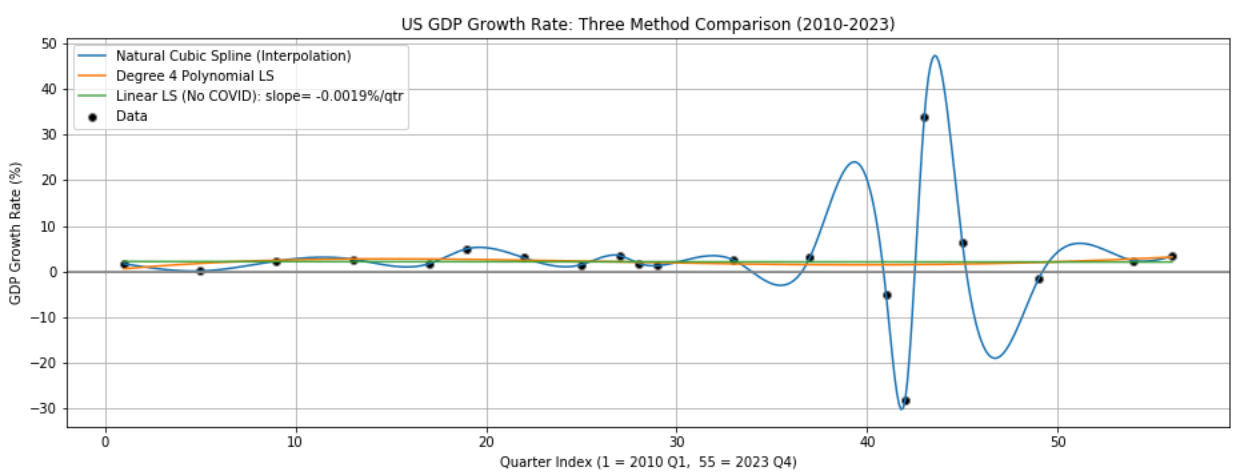

Figure 4. All three methods plotted: natural cubic spline (blue), degree-4 polynomial (orange), linear least squares without COVID-19 (green)

Big O notation analyzes the runtime of an algorithm and how it grows as the input size n grows. For example, O(n) means the memory usage changes linearly with n while $O(n^3)$ means if n doubled, usage would grow eight times as much. The dataset only has n = 20, and at such a small sample size, the runtime between algorithms is essentially identical as they all complete their programs within milliseconds. Although the polynomial and linear systems solve the same matrix regardless of n and the spline solves in $O(n^3)$, the difference is negligible. There is little to

Matthew Zeng
mz22353
March 31, 2026

no complexity difference, which means the metrics for determining the most optimal fit remains the same: smoothness, accuracy, and sensitivity to outliers.

Matthew Zeng
mz22353
March 31, 2026

CONCLUSION

For interpolation, the **natural cubic spline** is the most optimal fit. It exactly passes through all 20 data values, resulting in an RMSE = 0% while also pushing C^2 continuity onto each knot. However, its overarching weakness is its sensitivity to outliers and extreme values, which causes overshooting interpolations due to Runge's phenomenon. For trend analysis, the **linear least squares model (no COVID-19)** represented the pattern most accurately. An essentially zero (-0.0019%) slope per quarter reveals the stability of the US GDP growth of around 2% annually. It produced an RMSE = 1.44% compared to the degree-4 polynomial RMSE = 10.03%. These differences show that a more complicated model doesn't always mean a more accurate fit. When data is scattered and relatively even, a simpler fit may be more optimal. Ultimately, the decision should be mainly founded upon the desired use of the model. If the data accurately represents the intended process being analyzed, exact interpolation should be used. If the data raises questions or is distorted by external factors, inexact models should be used for approximation. By analyzing various statistical values, such as RMSE and residuals, we can determine the best option for each use.